PV and Battery Trading in MGP and MI Auctions
Current Mathematical Model and Execution Workflow

`bess_pv_trading` repository

Implementation reviewed on 20 October 2026

Abstract

This document describes the implemented optimization and execution system for a single photovoltaic farm and an AC-coupled battery sharing one point of interconnection. It distinguishes auction plans, immutable market commitments, physical feedback execution, and ex-post settlement. The shared Gurobi formulation supports a risk-neutral scenario tree. The current historical January experiment instead uses one deterministic rolling weather forecast and perfect auction-price information, followed by execution against measured AC production. Separate full-information PV+BESS and PV-only optimizations quantify an ex-post operating value of storage. Equations below follow the current source code; they are not a proposal or a claim of a fully causal historical trading strategy.

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1 Scope and implementation map

One farm has PV generation, an AC-coupled BESS, and a common AC grid connection. There is no local electricity demand. PV can be exported, stored, or curtailed; the battery can charge from PV or grid imports and discharge toward the grid. The market sequence is MGP (called DA in code), MI-A1, MI-A2, and MI-A3 (IDA1, IDA2, IDA3). Every submitted signed quantity is assumed to execute in full at the auction clearing price. Bid prices, acceptance uncertainty, market impact, and continuous intraday trading are outside this formulation.

The implementation has three distinct objects:

1. An optimization plan, which includes the current actionable auction order, future auction recourse, and a battery/PV schedule over the remaining horizon.
2. An immutable commitment ledger, updated at submission and publication.
3. A physical trajectory produced by a feedback/protection policy using realized interval-average PV. Its energy, wear, and deviations are settled separately from the optimization objective.

Source	Responsibility
`src/domain.py`	Validated grids, assets, scenarios, information nodes, states and solver contracts.
`src/optimization/model.py`	Physical, trading, product-resolution, optional imbalance constraints and objective.
`src/optimization/degradation.py`	Throughput coefficient and segmented cycle-depth model.
`src/optimization/validation.py`	Numerical validation independent of the solver constraint objects.
`src/transaction_fees.py`	Pure-Python fee integration, dated tariffs and operator monthly state.
`src/optimization/transaction_fees.py`	Flat-band LP costs and exact progressive ABS/PWL constraints.
`src/simulation/synthetic.py`	Shared event replay and auction submission/publication accounting.
`src/simulation/replay.py`	Residual scenario conditioning and bounded physical execution.
`src/simulation/gme_oracle.py`	Manual-data historical price oracle, rolling PV and full-information asset comparisons.
`src/simulation/settlement.py`	Exact-grid actual settlement and expected/realized KPI reconciliation.
`src/forecasting/pv.py`	Weather-to-AC PV conversion.
`src/markets.py`	Configurable local auction gates, publication events and DST-aware delivery grids.

2 Time, units, signs and information

2.1 Delivery intervals and auction products

A local delivery day is partitioned into physical intervals $\mathcal{T} = \{0, \dots, T-1\}$, with starts τ_t, ends τ_{t+1} and durations

$$\Delta_t = (\tau_{t+1} - \tau_t)/\text{one hour}. \tag{1}$$

The historical study uses quarter-hours, so $\Delta_t = 0.25$. A local day is constructed between successive midnights in `Europe/Rome`, then stored in UTC: T is 92, 96, or 100 for spring-DST, ordinary, or autumn-DST days. Repeated local hours are distinguished by their actual UTC offset/fold. No equation assumes that every day has exactly 24 elapsed hours.

A physical interval is not necessarily an independent market product. Before 1 October 2025, DA products are hourly; the MI auctions in the current 2025 history are already quarter-hourly. Physical battery controls are quarter-hourly at all four optimization stages. For the four equal-duration quarters of one native DA hour h, the implementation enforces

$$q_{\mathrm{DA},n,t} = q_{\mathrm{DA},n,t'} \quad (t, t' \in h). \tag{2}$$

These are equal MWh shares of one hourly order, whose total is $\sum_{t \in h} q_{\mathrm{DA},n,t}$. The hourly clearing price is repeated as an objective coefficient on those shares. The quarters do not become separate DA trades or separate observed prices. There is no corresponding date-based hourly equality on battery charging or discharging. Native product resolution and optional physical-control grouping are separate configuration fields.

The current study's configured local calendar is:

Auction	Gate	Results time used	Eligible delivery
DA	D−1 12:00	D−1 13:30	Whole day D
IDA1	D−1 15:00	D−1 15:30	Whole day D
IDA2	D−1 22:00	D−1 22:30	Whole day D
IDA3	D 10:00	D 10:30	D 12:00 onward

Submissions are planned five minutes before the gate in the supplied study. These times, date validity, result-time provenance and cancellations are explicit YAML settings. In particular, the historical DA publication time is a study assumption. Gate time and the start of IDA3 delivery are different events. Recorded publication timestamps can override configured times.

An auction a has an eligibility set $\mathcal{T}_a$. A documented/configured cancellation removes that auction from the active set. Missing data alone does not establish a cancellation: unexplained missing or incomplete curves fail validation. Absent auctions create no trade variables and no submission solve.

2.2 Notation and electrical basis

Symbol	Unit	Meaning
E	MWh	Nominal internal stored-energy capacity.
$E^{\min}, E^{\max}$	MWh	Operational energy limits derived from SOC percentages.
$\overline{P}$	MW AC	One symmetric battery charging/discharging rating.
$P^{\mathrm{imp}}, P^{\mathrm{exp}}$	MW AC	Directional shared POI ratings.
η_c, η_d	fraction	Complete AC-bus-to-storage and storage-to-AC efficiencies.
σ	fraction/hour	Configured hourly self-discharge fraction.
$V_{s,t}$	MW AC	Forecast available PV in scenario s.
$\widetilde{V}_t$	MW AC	Effective realized PV used for execution.
$c_{n,t}, d_{n,t}$	MW AC	Battery charge/discharge commands at information node n.
$u_{n,t}$	MW AC	PV curtailment.

Symbol	Unit	Meaning
$i_{s,t}$	MW AC	Uncurtailed PV injection into the common AC bus.
$g_{s,t}$	MW AC	Net grid exchange; positive is export, negative is import.
$e_{s,t}$	MWh	Internal battery energy at an interval boundary.
$q_{a,n,t}$	MWh	Signed auction adjustment; positive sale, negative purchase.
$Q_{s,t}$	MWh	Final cumulative contracted delivery position.
$\pi_{a,s,t}$	EUR/MWh	Auction-price coefficient for interval t.
w_s	fraction	Scenario probability, summing to one.

Power variables are AC at the common bus/POI. Battery energy is internal stored energy. Nameplate PV capacity is DC; measured historical `power` is already AC kW, converted once to MW. A separate PV-output/transformer conversion is used only when predicting AC PV from modeled DC. BESS efficiency is not applied to measured PV or direct PV exports.

2.3 Forecast vintages and observability

A weather forecast cycle v has source time t_v and availability time A_v. Without explicit source publication times, the configured convention is $A_v = t_v + 4$ hours. At decision cutoff k, the rolling policy chooses the latest eligible cycle with $A_v \leq k$ and required remaining-horizon coverage. Weather cycles about six hours apart are successive forecast vintages, not ensemble members. The current dataset supplies one deterministic weather member.

Measured interval-average PV becomes available after interval end plus configured telemetry delay. A forecast for future intervals is not replaced by future realized power in the ordinary auction replay. If a decision occurs inside an unfinished quarter-hour, the next boundary state is projected through that interval with forecast PV and the same execution policy. Its final realized average cannot yet be observed. The hindsight counterfactuals deliberately have future PV and price information and are labeled as such.

Policy-treated observations carry evidence times. Plateau evidence can require later samples; two-sided interpolation requires its right endpoint and applicable mask evidence. A finalized corrected time series therefore differs from the as-of view available to a past decision. This remains relevant even when a study uses actual-price foresight.

3 PV power from weather forecasts

3.1 Irradiance and orientation

The input mappings declare forecast shortwave radiation as GHI and direct radiation as horizontal beam irradiance (BHI), with the latter an explicit working assumption. Hourly radiation means are end-labelled and redistributed uniformly over contained quarters. Other fields follow their declared instantaneous/accumulation semantics rather than the same blanket interpolation. Solar geometry is evaluated at each quarter-hour midpoint using configured coordinates and farm elevation.

For eligible irradiance data,

$$G_{\mathrm{DHI}} = G_{\mathrm{GHI}} - G_{\mathrm{BHI}}, \qquad G_{\mathrm{DNI}} = \frac{G_{\mathrm{BHI}}}{\cos\theta_z} \quad \text{away from the low-sun cutoff.} \tag{3}$$

The code uses pvlib's DNI helper with configured low-sun handling and a finite DNI cap or invalidation rule. BHI exceeding GHI is either invalidated or explicitly clipped and flagged. Optional

Erbs decomposition can replace missing/unusable direct input under its declared conditions; it is not a general repair of contradictory source values.

Fixed arrays use metadata tilt and a declared source-to-pvlib azimuth conversion. Tracking IDs 1 and 3 use `pvlib.tracking.singleaxis` with configurable axis tilt, axis azimuth, rotation limit, backtracking and ground-coverage ratio. Type 3 currently requires the explicit front-side-only approximation; rear-side bifacial gain is omitted. The current geometry uses an acknowledged Milan location proxy rather than verified farm coordinates.

With panel tilt β, incidence angle α and ground albedo a, the isotropic transposition can be written

$$G_{\mathrm{POA}} = G_{\mathrm{DNI}} \left[\cos \alpha\right]_{+} + G_{\mathrm{DHI}} \frac{1 + \cos \beta}{2} + a G_{\mathrm{GHI}} \frac{1 - \cos \beta}{2}. \tag{4}$$

3.2 Temperature, DC output and AC conversion

Wind speed is the magnitude of the configured horizontal wind components, $v = \sqrt{v_u^2 + v_v^2}$. The Faiman temperature model is

$$T_{\mathrm{mod}} = T_{\mathrm{air}} + \frac{G_{\mathrm{POA}}}{U_0 + U_1 v}. \tag{5}$$

The supplied wind height is explicitly acknowledged as a module-wind proxy. The resulting temperature is passed to pvlib's PVWatts DC calculation:

$$P_{\mathrm{DC}} = \max \left\{ 0,\ P_{\mathrm{DC,STC}} \frac{G_{\mathrm{POA}}}{1000} \left[1 + \gamma (T_{\mathrm{mod}} - T_{\mathrm{ref}})\right] \right\}. \tag{6}$$

DC power is scaled to AC by the configured explicit PV-output efficiency η_{PV}, or by an explicitly fitted DC-to-AC coefficient κ:

$$\widehat{P}_{\mathrm{AC}} = \begin{cases} \eta_{\mathrm{tr}} \eta_{\mathrm{PV}} P_{\mathrm{DC}}, & \text{explicit-efficiency route,} \\ \eta_{\mathrm{tr}} \kappa P_{\mathrm{DC}}, & \text{fitted-coefficient route.} \end{cases} \tag{7}$$

A configured AC cap clips this output if present; the night mask and validity flags are then applied. The present study uses an unfitted explicit output factor and a configurable separate transformer factor. This is not a full calibrated module-and-inverter electrical model. Temperature and wind affect production through temperature; irradiance and orientation affect POA. Cloud/dew-point fields retained in the store are not additional direct PV predictors here.

Invalid required inputs remain invalid unless an explicit policy applies. Historical GME studies may use their configured night-zero forecast policy and a bounded fallback for missing realized-power intervals. Both the number and identity of substituted intervals are recorded. All counterfactuals use the same *effective* realized PV series, including these substitutions.

4 Optimization at an auction decision

Each residual solve uses only the remaining suffix of the delivery grid. In the equations below that suffix is reindexed from zero, with T denoting its final boundary; its original UTC product identifiers are retained for commitments.

4.1 Scenario tree and nonanticipativity

At decision k, let $\mathcal{S}_k$ be represented leaf scenarios and let $\mathcal{A}_k$ be the open active auctions. An information tree assigns an auction bid owner $n_a(s)$ to each auction/leaf and a control owner $n_t(s)$

to each interval/leaf. Curtailment may have a separate owner $m_t(s)$ under the selected execution-information convention.

Decisions are node-owned variables, not independent leaf decisions followed by a penalty for disagreement. For example,

$$c_{s,t} = c_{n_t(s),t}, \quad d_{s,t} = d_{n_t(s),t}, \quad q_{a,s,t} = q_{a,n_a(s),t}, \quad u_{s,t} = u_{m_t(s),t}. \tag{8}$$

Thus two scenarios with the same observable history share actionable decisions. A latent PV-member label is not available information by itself. Constraints apply to every represented leaf, including zero-probability leaves if supplied; probabilities only weight the objective. In the current one-scenario historical price oracle, the tree degenerates to deterministic planning with known future prices and a forecast PV trajectory.

4.2 Physical variables and balances

For every scenario and remaining interval:

$$i_{s,t} + u_{s,t} = V_{s,t}, \qquad i_{s,t} \geq 0, \qquad 0 \leq u_{s,t} \leq V_{s,t}, \tag{9}$$

$$g_{s,t} = i_{s,t} + d_{s,t} - c_{s,t}, \tag{10}$$

$$-P^{\text{imp}} \leq g_{s,t} \leq P^{\text{exp}}. \tag{11}$$

Curtailment appears only in the PV balance. It cannot erase bought grid energy or battery charging energy. A negative grid exchange must be absorbed by net battery charging since there is no local demand. The model does not explicitly assign charging power to its PV/grid source; that partition is an accounting convention used in post-processing.

With $z_{n,t} \in \{0, 1\}$ denoting charging mode,

$$0 \leq c_{n,t} \leq \overline{P} z_{n,t}, \qquad 0 \leq d_{n,t} \leq \overline{P}(1 - z_{n,t}). \tag{12}$$

This excludes simultaneous charge and discharge, uses one symmetric AC rating, and introduces no separate battery-inverter power limit.

SOC percentage parameters $s_{\min}, s_{\max}$ define

$$E^{\min} = E\frac{s_{\min}}{100}, \qquad E^{\max} = E\frac{s_{\max}}{100}, \qquad E^{\min} \leq e_{s,t} \leq E^{\max}. \tag{13}$$

For $\rho_t = (1 - \sigma)^{\Delta_t}$, the transition is

$$e_{s,t+1} = \rho_t e_{s,t} + \eta_c c_{s,t} \Delta_t - \frac{d_{s,t}\Delta_t}{\eta_d}. \tag{14}$$

One AC MWh charged stores η_c MWh. Delivering one AC MWh consumes $1/\eta_d$ internal MWh. In combined mode the two configured efficiencies each cover their entire directional path. In converter-and-cell mode, $\eta_c = \eta_{c,\text{conv}}\eta_{c,\text{cell}}$ and $\eta_d = \eta_{d,\text{conv}}\eta_{d,\text{cell}}$. The alternatives cannot be mixed. The round-trip factor is $\eta_c\eta_d$. Self-discharge acts on stored energy, not on a power variable, and is separate from cycle degradation cost.

The initial boundary is fixed to measured/carried/projected state e^k:

$$e_{s,0} = e^k. \tag{15}$$

If the terminal target is configured, also impose

$$e_{s,T} = e^{\text{target}}. \tag{16}$$

In the GME adapter, initial/terminal percentages are converted using nominal capacity. A `null` terminal SOC omits (16). Prepared and synthetic boundary-state contracts use explicitly named MWh values. Neither version adds a terminal energy salvage value to the objective.

4.3 Auction quantities and physical delivery

Let Q_t^{fix} contain all previously submitted commitments, priced or still pending. The new final position on leaf s is

$$Q_{s,t} = Q_t^{\text{fix}} + \sum_{a \in \mathcal{A}_k :\, t \in \mathcal{T}_a} q_{a,n_a(s),t}. \tag{17}$$

An ineligible auction/interval has no trade variable. The model allows signed sales/purchases; they are adjustments to prior positions, not gross generation. For every eligible trade,

$$-P^{\text{imp}} \Delta_t \le q_{a,n,t} \le P^{\text{exp}} \Delta_t. \tag{18}$$

For every open auction a in chronological order, define the cumulative planned position after that auction as

$$C_{s,t}^{a} = Q_t^{\text{fix}} + \sum_{b \in \mathcal{A}_k :\, b \le a,\, t \in \mathcal{T}_b} q_{b,n_b(s),t}. \tag{19}$$

It satisfies

$$-P^{\text{imp}} \Delta_t \le C_{s,t}^{a} \le P^{\text{exp}} \Delta_t, \qquad -P^{\text{imp}} \Delta_t \le Q_{s,t} \le P^{\text{exp}} \Delta_t. \tag{20}$$

There is no independent commercial volume cap. The directional POI ratings apply to individual auction adjustments, cumulative positions after each auction, and physical delivery. An offsetting later trade is allowed, subject to its own POI-derived limit and the resulting position limit. Thus the model permits bounded cross-auction spread trading even without a battery; its revenues can be present in both asset counterfactuals.

With `trading.optimize_imbalances: false`, which is the current farm-910 policy, planned delivery is exact in every interval:

$$g_{s,t} \Delta_t = Q_{s,t}. \tag{21}$$

No positive/negative imbalance variables or imbalance direction binary are created. Forecast deviations are not an additional dispatch resource. An optional configured product-group delivery equality can additionally impose $\sum_{t \in h} (g_{s,t} \Delta_t - Q_{s,t}) = 0$. It is redundant with (21) and does not replace it in the disabled-imbalance mode.

4.4 Current objective and sunk cash flows

For throughput degradation with coefficient c_{thr} and an optional import fee f, the current residual objective is

$$\max J_k = \sum_{s \in \mathcal{S}_k} w_s \left[\sum_{a \in \mathcal{A}_k} \sum_{t \in \mathcal{T}_a} \pi_{a,s,t} q_{a,n_a(s),t} - c_{\text{thr}} \sum_t (c_{s,t} + d_{s,t}) \Delta_t - f \sum_t \left[-g_{s,t} \right]_+ \Delta_t - C_s^{\text{trading}} \right]. \tag{22}$$

The implementation uses an exact import-power linearization with a direction binary only when f is nonzero. The farm-910 study sets $f = 0$.

4.5 Market transaction fees and operator monthly volume

The optional `transaction_fees.mode` is `none` (default), `flat`, or `gme_real`. Let $m(a)$ be the calendar month of auction submission in market local time, V_m known already executed operator volume, and $G_{m,s} = \sum_{a : m(a) = m,\, t} |q_{a,n_a(s),t}|$ new gross MWh. Purchases and sales count separately even when final delivery is zero. Quarter-hour energy shares of an hourly product are summed once, without another duration factor. For a cumulative progressive tariff F_m,

$$C_s^{\text{trading}} = \sum_m \left[F_m(V_m + G_{m,s}) - F_m(V_m) \right]. \tag{23}$$

In flat mode this is the configured EUR/MWh coefficient times gross MWh. The published GME MPE marginal rates are 0 on the first 20,000 MWh/month, 0.04 up to 1,000,000, 0.03 up to 10,000,000, and 0.02 thereafter. These bands are progressive and not a retroactive rate on the monthly total.

Fee-free horizons introduce no auxiliary variables. Positive single-band costs use linear absolute-value epigraphs. Across reachable breakpoints, exact Gurobi ABS general constraints and a PWL equality represent both gross volume and the nonconvex tariff; a convex epigraph alone is not used for regressive bands. Absolute-value variables retain auction-node ownership and native-product sharing. Costs are evaluated for each leaf and weighted by w_s.

The fee state includes pending guaranteed transactions immediately at submission. Publication adds no second fee. Executed gross volume remains in V_m during residual solves, while incurred EUR are sunk. The state carries an availability clock and declared provenance; nonzero future executed monthly volume is rejected. Months use transaction dates, so pre-DA/IDA1/IDA2 and IDA3 for one delivery day may belong to different months. A new month starts at zero. The saved final fee state can initialize a later chronological study. Operator scope must explicitly distinguish a standalone plant from a portfolio with exogenous starting volume. Unknown future external activity is not invented: costs are exact conditional on the supplied initial state, not a complete operator-level monthly forecast.

Tariffs have explicit nonoverlapping complete-month validity windows, version, source, thresholds, marginal rates and an applicability status. The current GME tariff and the 2006 annual report show the same structure, the latter effective in 2007. A dated confirmation of 2025 applicability was not established; the supplied historical example marks this explicitly as `assumed_unchanged`. Unsupported dates fail. Example YAMLs are `configs/full_experiments/synthetic_fees_none.yaml`, `configs/full_experiments/synthetic_fees_flat.yaml`, and `configs/full_experiments/synthetic_fees_gme.yaml`. Membership, taxes, ACER record charges and separate aggregator commissions remain excluded; physical grid fees and degradation remain separate cost components.

Past auction cash flow and already incurred degradation are sunk at residual reoptimization. Priced commitments remain in the external ledger, and pending unpriced submitted quantities enter Q^{fix} without adding an optimizable cash-flow term. Their future clearing value is constant with respect to all remaining decisions and is excluded from J_k. This avoids double counting. The initial pre-DA solve has no prior commitments; its objective spans the modeled whole-day opportunity set. Residual objective values therefore should not be summed across auction solves to compute daily profit.

There is no explicit curtailment penalty, risk measure, capital cost, fixed O&M, tax, brokerage fee, ramp constraint, minimum dwell time, or calendar-aging charge in (22). Curtailment can affect revenue through delivery choices, but is not independently priced. The horizon can exploit inventory when its terminal target is free; this matters when comparing asset value.

4.6 What is free at each of the four decisions

Decision	New free auction quantities	Previously submitted fixed quantities
DA	DA, IDA1, IDA2, IDA3 plans	None
IDA1	IDA1, IDA2, IDA3 plans	DA
IDA2	IDA2, IDA3 plans	DA and IDA1
IDA3	IDA3 eligible products	DA, IDA1 and IDA2

At each decision, remaining-horizon charge/discharge modes, curtailment, grid powers and energy states are free subject to their balances, boundary state, nonanticipativity, product rules and

terminal condition. Executed past intervals are absent from the residual grid. Future-auction quantities are plans until their own submission. Only the current auction order is committed from that solve. Before noon, IDA3 cannot change orders for the morning; physical control replanning of remaining earlier intervals must respect their already fixed market positions.

5 Battery degradation models

5.1 Current AC throughput approximation

For one trajectory,

$$H = \sum_t (c_t + d_t)\Delta_t, \qquad D_{\text{thr}} = c_{\text{thr}} H. \tag{24}$$

Both directional AC throughputs are charged once. The YAML can directly supply c_{thr} or derive it from replacement economics and reference cycle life. Let p_{USD} be pack USD/kWh, x be USD per EUR, p_{extra} additional EUR/kWh replacement cost, s_R salvage fraction, N reference cycle life and d_{ref} reference internal depth of discharge. Then

$$R = 1000E\left(\frac{p_{\text{USD}}}{x} + p_{\text{extra}}\right)(1 - s_R), \tag{25}$$

$$H_{\text{life}} = N d_{\text{ref}} E\left(\frac{1}{\eta_c} + \eta_d\right), \tag{26}$$

$$c_{\text{thr}} = \frac{R}{H_{\text{life}}}. \tag{27}$$

One reference internal cycle charges $d_{\text{ref}}E/\eta_c$ AC MWh and discharges $d_{\text{ref}}E\eta_d$ AC MWh, explaining the normalization. There is no additional factor of two or half applied to (27). The configured end-of-life capacity fraction is descriptive metadata, not an extra multiplier or a dynamic capacity update.

The current inherited farm-910 YAML uses a historical generic pack-price proxy, a dated FX conversion, and an illustrative reference cycle life. Those are study inputs with source metadata, not an estimate of a specific installed BESS cost. The resolved R and c_{thr} are saved. The cost is financial wear; it does not reduce E, $\overline{P}$, or efficiencies as the simulation proceeds.

5.2 Optional segmented depth-dependent aging

The optional `xu_segmented` mode follows the configured segmented cycle-depth construction described in the repository's Xu notes and the author paper [1]. It is distinct from the current throughput study. For a configurable convex stress curve $\Phi(d) = \alpha d^\beta$, $\beta \geq 1$, partition nominal internal capacity into J equal segments of size E/J. The marginal AC discharge cost of segment $j = 1, \ldots, J$ is

$$m_j = \frac{RJ}{\eta_d E}\left[\Phi(j/J) - \Phi((j-1)/J)\right]. \tag{28}$$

It is nondecreasing with depth. The degradation charge is

$$D_{\text{Xu}} = \sum_t \sum_{j=1}^{J} m_j d_{j,t} \Delta_t. \tag{29}$$

Only discharging is charged under this convention. A discharging half-cycle incurs the convention's depth cost, while a charging half-cycle does not; do not add an extra factor of one half. Nonzero throughput cost is rejected in this mode.

Segment flows and energies obey

$$\sum_j c_{j,t} = c_t, \qquad \sum_j d_{j,t} = d_t, \qquad \sum_j h_{j,t} = e_t, \tag{30}$$

$$h_{j,t+1} = h_{j,t} + \Delta_t(\eta_c c_{j,t} - d_{j,t}/\eta_d), \qquad 0 \le h_{j,t} \le E/J. \tag{31}$$

Segment energies represent cycle-depth memory, not fixed absolute SOC buckets. Charging and discharging both use shallow available segments first. For each $j < J$, the implementation adds priorities $v^c_{j,t}, v^d_{j,t} \in \{0, 1\}$:

$$\sum_{\ell>j} c_{\ell,t} \le \overline{P} v^c_{j,t}, \qquad h_{j,t+1} \ge (E/J) v^c_{j,t}, \tag{32}$$

$$\sum_{\ell>j} d_{\ell,t} \le \overline{P} v^d_{j,t}, \qquad h_{j,t+1} \le (E/J)(1 - v^d_{j,t}). \tag{33}$$

Using a deeper segment therefore requires the shallower segment to end full when charging or empty when discharging. This chooses a deterministic shallow- first representative, including otherwise indifferent charging allocations. Segment actions have the same information ownership as aggregate battery power.

At a declared fresh study start, initialize

$$h_{j,0} = \mathrm{clip}\left(e_0 - (j-1)E/J,\ 0,\ E/J\right). \tag{34}$$

Afterwards the segment vector must be carried with actual execution; an arbitrary residual solve cannot reconstruct it from aggregate SOC. Previous throughput or aging requires explicit memory. Xu mode currently rejects nonzero self-discharge. It remains a finite-segment approximation and omits calendar, temperature and current-rate aging, capacity-fade feedback and unknown pre-study cycle history.

6 Submission, publication and reoptimization workflow

Before each active auction submission, the GME replay refreshes the latest available PV forecast over unexecuted intervals, projects a boundary if necessary, and solves the residual planning problem. Previously submitted quantities, including still-unpriced ones, are fixed. At submission, only the current auction's node-shared action is extracted. Later planned auction quantities are not booked and can change.

At publication, known prices are updated, and actual contractual revenue $\pi^{\mathrm{real}}_{a,t} q^{\mathrm{sub}}_{a,t}$ is entered exactly once in the ledger. Publication does not reopen a market. The current GME experiment forces `reoptimize_physical_controls=false`; it uses four auction solves for four active auctions, not an extra set of publication solves or 96 physical optimizations. Physical feedback is algebraic. Synthetic replay can separately enable interval-control reoptimization, so its solve count need not be four.

At the next auction, closed commitments are carried, remaining open opportunities are reconsidered, and past executed state/aging remain fixed. In the actual-price oracle all future auction prices are already supplied as a benchmark, so price publication does not remove price uncertainty there. In a forecast-based route, known results condition subsequent price distributions without changing the commitments that produced them.

7 Physical feedback with realized PV

7.1 Feasible interval power range

The current policy is `track_committed_grid`. For interval t its target is

$$g_t^\star = Q_t^{\text{committed}}/\Delta_t. \tag{35}$$

The controller uses effective realized interval-average PV $\widetilde{V}_t$; this is an idealized interval simulator, not a subsecond telemetry/control model. It does not disclose this final average to the earlier auction optimizer. Given actual initial energy e_t and retained energy $\bar{e}_t = \rho_t e_t$, the feasible charging/discharging maxima are

$$C_t = \min\left\{\overline{P}, \frac{[E^{\max} - \bar{e}_t]_+}{\eta_c \Delta_t}\right\}, \tag{36}$$

$$D_t = \min\left\{\overline{P}, \frac{\eta_d \left[\bar{e}_t - E^{\min}\right]_+}{\Delta_t}\right\}. \tag{37}$$

First clip the planned charge/discharge commands to these maxima, obtaining c_t^0, d_t^0. Simultaneous positive requested charging and discharging after protection is rejected. Initially curtail only available PV: $u_t^0 = \min\{\widetilde{V}_t, \left[u_t^{\text{plan}}\right]_+\}$.

7.2 Grid-tracking policy

Release scheduled curtailment if it can help meet a deficit:

$$g_t^0 = \widetilde{V}_t - u_t^0 + d_t^0 - c_t^0, \tag{38}$$

$$\bar{u}_t = u_t^0 - \min\{u_t^0, \left[g_t^\star - g_t^0\right]_+\}. \tag{39}$$

The desired net battery dispatch and its feasible projection are

$$b_t^{\text{req}} = g_t^\star - \widetilde{V}_t + \bar{u}_t, \tag{40}$$

$$b_t = \text{clip}(b_t^{\text{req}}, -C_t, D_t), \tag{41}$$

$$c_t = [-b_t]_+, \qquad d_t = [b_t]_+. \tag{42}$$

Finally curtail any remaining surplus that battery adjustment cannot absorb:

$$u_t = \bar{u}_t + \min\{\widetilde{V}_t - \bar{u}_t, \left[\widetilde{V}_t - \bar{u}_t + b_t - g_t^\star\right]_+\}. \tag{43}$$

Thus excess PV first decreases discharge, then increases charging as needed and feasible, then is curtailed. A PV shortfall first releases planned curtailment, then reduces charging or increases discharge within the actual SOC box. The remaining tracking error is preserved if battery power/energy cannot supply it.

POI protection follows. If net import exceeds its rating, charging is reduced. If export exceeds its rating, discharge is reduced first and remaining excess PV is curtailed. Grid power is recomputed consistently with the adjusted actions. Actual energy then advances using (14) with these executed powers, and bounds are checked before accepting the new state. Both throughput/Xu memory and cumulative financial wear are advanced from executed actions, not from the old auction plan. Protection is mandatory under either execution policy; the alternative `follow_battery_plan` skips grid-tracking adjustment.

There is no controller constraint forcing the actual terminal energy to the auction plan's target. Forecast errors and grid tracking can cause an ending-SOC difference even if every optimization imposed (16). This is visible in the current terminal-50% study and is not equivalent to guaranteeing 50% realized SOC. Any hard actual-terminal policy would be an additional controller design and could create different imbalances/cash flows.

7.3 PV source allocation for reporting

For the executed trajectory, use the explicit PV-first accounting convention

$$P_t^{\mathrm{PV}\to\mathrm{B}} = \min\{\widetilde{V}_t - u_t, c_t\}, \tag{44}$$

$$P_t^{\mathrm{PV}\to\mathrm{G}} = \widetilde{V}_t - u_t - P_t^{\mathrm{PV}\to\mathrm{B}}, \tag{45}$$

$$P_t^{\mathrm{PV}\to\mathrm{curt}} = u_t. \tag{46}$$

They sum to $\widetilde{V}_t$ in every interval. Grid-supplied charging and battery discharge are not counted as PV production. These are post-processing flows; there are no separate source-attribution decision variables in the MILP.

8 Settlement and expected-versus-realized accounting

Actual net deviation is retained at the native quarter-hour:

$$\delta_t^{\mathrm{real}} = g_t^{\mathrm{real}}\Delta_t - Q_t^{\mathrm{committed}}. \tag{47}$$

Positive means more net export than promised. With the selected single Terna NORD settlement price λ_t, the modeled cash flow is

$$S_t = \lambda_t \delta_t^{\mathrm{real}}. \tag{48}$$

A positive price therefore does not mean every imbalance is penalized: positive excess export is credited and a negative deviation is debited. Negative prices are allowed and reverse these cash-flow signs. The general settlement helper also accepts two directional curves:

$$S_t = \lambda_t^{+}\left[\delta_t^{\mathrm{real}}\right]_+ - \lambda_t^{-}\left[-\delta_t^{\mathrm{real}}\right]_+. \tag{49}$$

This documents the implemented study accounting, not every possible Italian participant-specific settlement item or regulatory regime. The workbook adapter selects the declared NORD macrozone and the imbalance-price column, retaining other imported fields as provenance. Product starts and ends must match the execution grid exactly. Prices are neither hourly-averaged nor filled across gaps; missing enabled settlement prices are an error.

Actual daily auction cash flow and settled operating profit are

$$A^{\mathrm{real}} = \sum_{a,t} \pi_{a,t}^{\mathrm{real}} q_{a,t}^{\mathrm{sub}}, \tag{50}$$

$$J^{\mathrm{real}} = A^{\mathrm{real}} + \sum_t S_t - D^{\mathrm{executed}} - f\sum_t \left[-g_t^{\mathrm{real}}\right]_+ \Delta_t - C^{\mathrm{trading,real}}. \tag{51}$$

A model deterrence penalty is not an actual settlement charge and is not subtracted a second time by actual post-processing. If settlement is unavailable, profit before settlement is reported and fully settled profit remains unknown. Only actually submitted guaranteed transactions enter $C^{\mathrm{trading,real}}$; future plans that are never submitted contribute no realized fees. The ledger retains transaction-month volume, per-trade charges and tariff versions. An independent Python calculation reconciles those charges and the final monthly state. Expected and realized transaction fees are reported separately from gross auction revenue. In `realized_profit_breakdown.png`, trading-fee EUR and gross MWh occupy the former import-fee slot; nonzero physical grid fees appear as their own additional bar.

Let J^{init} and A^{init} be the initial auction solve's expected profit and auction revenue. Saved mismatches are

$$\Delta J = J^{\mathrm{real}} - J^{\mathrm{init}}, \qquad \Delta A = A^{\mathrm{real}} - A^{\mathrm{init}}, \qquad r_J = 100\frac{\Delta J}{|J^{\mathrm{init}}|}. \tag{52}$$

The percentage is undefined/`null` when its expectation is zero; unsettled profit comparisons remain unknown. Differences include later replanning and physical execution effects, not only price-forecast error.

The replay also evaluates each stored auction snapshot as an open-loop diagnostic against realized PV and the same settlement prices. Its planned battery commands are protected physically. Deviations are measured against that snapshot's planned grid schedule (which can include assumed future trades) or the final contract, as labeled in the recorded comparisons. It is not another auction optimization. Because IDA3 covers a shorter residual horizon, a common-horizon comparison is also saved; unaligned whole-horizon totals should not be compared as if their delivery intervals were identical.

9 Perfect-information schedules and the PV-only baseline

9.1 PV+BESS versus the feedback controller

After replay, one additional full-day problem uses $V_t = \widetilde{V}_t$ and all actual auction-price curves, with a single scenario and all auction orders jointly free subject to their native products, eligibility, POI and physical constraints. It retains the BESS ratings, efficiencies, SOC box, degradation, initial energy and configured terminal target. Call its achieved solver objective $J^{\mathrm{PI}}_{\mathrm{PV+BESS}}$.

This is deliberately a hindsight plan rather than a causally executable policy. It is compared with the feedback controller's actual dispatch, curtailment and SOC, and its objective is saved as the solver's `ObjVal`. Four missing measurement intervals, if permitted by the selected run, are handled through that run's explicit fallback; diagnostics record their count and identities. The schedule figure shows both trajectories and their battery-power difference through the final day boundary. It does not assert that feedback should reproduce the hindsight trajectory or terminal energy.

9.2 PV-only asset counterfactual

When `experiment_outputs.pv_only_baseline: true`, a second independent solve uses the same optimizer builder with

$$E = 0, \quad \overline{P} = 0, \quad E^{\min} = E^{\max} = 0, \quad e_0 = e_T = 0, \quad D = 0. \tag{53}$$

Battery powers and energy variables are consequently fixed to zero; the code reuses one physical/trading formulation instead of implementing a second, potentially inconsistent PV-only model. Unused variables can be removed by solver presolve. PV injection/curtailment and auction trades remain free subject to the same effective realized PV, native products, active auctions and POI. Call its achieved objective $J^{\mathrm{PI}}_{\mathrm{PV}}$.

The incremental operating-value KPI is

$$\Delta J^{\mathrm{PI}}_{\mathrm{BESS}} = J^{\mathrm{PI}}_{\mathrm{PV+BESS}} - J^{\mathrm{PI}}_{\mathrm{PV}}. \tag{54}$$

The BESS side includes its own modeled degradation; the PV-only side has no battery wear. Both sides retain permitted cross-auction spread trading, so market-trading revenue common to both is not credited to battery ownership. The KPI does not subtract realized feedback wear from a different ideal schedule. Each counterfactual reports its own objective components. Directly subtracting a PV-only hindsight objective from the realized PV+BESS feedback profit would mix information and execution policies and is not the asset-value KPI.

This comparison is one-day operating value before capital expenditure, fixed operating costs, financing and taxes. A free terminal target can also use initial stored energy or leave different

inventory; the selected 50% initial/terminal hindsight comparison avoids that particular inventory difference for the BESS model. Realized feedback may still finish at a different SOC. No residual-energy valuation, annualization, payback, NPV or investment optimization is implemented. Those need a representative horizon and additional cost/cash-flow assumptions.

Setting the switch to `false` skips the PV-only solve and its model, JSON, CSV and figure. Incremental-value metrics become `null` and an explicit baseline-enabled KPI is false. The perfect-information PV+BESS-versus-feedback comparison still runs. Four active auctions therefore produce four replay solves plus one PV+BESS hindsight solve and, when enabled, one PV-only solve. Counterfactual solve diagnostics are separate from replay solve counts.

10 Forecast prices and the general stochastic route

The historical GME study described above has no price noise and uses actual prices in every auction decision. The repository separately implements chronological point-price studies and an artifact-to-optimization route for uncertain prices and probabilistic PV members.

A generic selected linear model is $\widehat{\pi} = x^\top \widehat{\beta}$, possibly modeling a change from yesterday's price rather than the absolute level. Calendar, lagged prices, daily profiles and configured regional load/renewable forecasts form features. OLS, ridge, weighted ridge, lasso, elastic net and Huber estimators share training-only preprocessing and portable fitted artifacts. Validation selects specifications before later calibration and evaluation.

For an auction's native delivery vector, Gaussian calibration uses out-of-sample errors to estimate a bias b_a and temporal covariance Σ_a. Paths are

$$\pi_a = \mu_a(\mathcal{H}_a) + b_a + \varepsilon_a, \qquad \varepsilon_a \sim \mathcal{N}(0, \Sigma_a). \tag{55}$$

An example covariance policy is

$$\Sigma_a = (1-\gamma)S_a + \gamma \operatorname{diag}(S_a) + \epsilon I, \tag{56}$$

with saved shrinkage and variance-floor settings. Bias and zero-mean innovations are distinct; finite Monte Carlo samples need not have exactly zero sample mean. IDA conditional means may be reconstructed from a predecessor-price curve plus a learned spread, using only ancestor sampled/known clearing curves at their information nodes. Innovations are temporally correlated within an auction equation and independent across auction equations; their conditional means propagate inter-auction dependence. Negative simulated prices are not clipped.

For B branches at each of K unknown auction results there are up to B^K price paths per supplied PV member, before history merging and configured limits. Two branches at four unknown auctions give up to 16 price paths per member. Known results are fixed, not sampled. The prepared route forms joint scenario mass from PV-member and price-path masses; price draws shared across members represent the declared independent product-measure assumption rather than estimated PV/price dependence. Member identity remains latent until observations distinguish histories. The actual-price oracle repeats a common price curve across members and adds no artificial price noise.

Hourly and quarter-hour price regimes and DST templates remain distinct. Historical retrieved revisions lacking publication evidence are marked retrospective. A model fit cutoff, static feature availability, forecast cycle, and source timestamp must satisfy the selected decision cutoff policy. Explicit retrospective opt-in does not establish an operationally causal backtest. The prepared examples need their complete supplied tables/artifacts; they are not a fully populated alternative historical strategy by default.

11 Optional optimized-imbalance formulation

This section describes supported code that is disabled in the current farm-910 study. When `optimize_imbalances=true`, replace (21) by

$$\delta^+_{s,t} - \delta^-_{s,t} = g_{s,t}\Delta_t - Q_{s,t}, \tag{57}$$

$$0 \le \delta^+_{s,t} \le M_t y_{s,t}, \tag{58}$$

$$0 \le \delta^-_{s,t} \le M_t(1 - y_{s,t}), \qquad y_{s,t} \in \{0,1\}. \tag{59}$$

Here $M_t = (P^{\text{imp}}+P^{\text{exp}})\Delta_t$, optionally reduced by a configured imbalance-energy cap. The common M_t is valid because physical and contractual net energy use the same directional POI range. The sign binary prevents simultaneous positive/negative deviations, including with negative prices. Add to each scenario's objective

$$\sum_t \left[\lambda^+_{s,t}\delta^+_{s,t} - \lambda^-_{s,t}\delta^-_{s,t} - \kappa(\delta^+_{s,t} + \delta^-_{s,t})\right]. \tag{60}$$

If this mode has zero settlement price, zero penalty and no useful delivery restriction, an optimized contract need not be backed by delivered energy. That is why the present experiment disables optimized deviations and computes physical mismatches only afterwards. The optional formulation is a mathematical study mode, not an assertion that deliberate imbalance strategies are permitted.

12 Data provenance, outputs and reproducibility

Farm inputs are local read-only Zarr/Parquet stores. API and workbook records use `data/api/market.duckdb`. Raw/normalized snapshots, source identities, checksums, revision and retrieval metadata support repeated offline studies. Historical publication remains a separate field/convention. Provider acquisition is an explicit user-run command; neither the MILP builder nor the historical experiment downloads missing inputs. DuckDB writers and analysis processes must be coordinated; complete cached days can be studied before an entire acquisition window is complete.

YAML controls study date, calendar validity, source mappings, PV and data policies, BESS and POI characteristics, state, degradation, solver and outputs. One include is resolved relative to the containing YAML, with recursive mapping overrides. The current 50% study is configured by `configs/full_experiments/gme_oracle_2025-01-01_farm910_perfect_pv_comparison_terminal_soc_50.yaml`. Its inherited assumptions include a 5 MWh battery, 2.5 MW symmetric power, 10–90% SOC box, 0.97 combined efficiencies and a 6.638 MW directional POI. These are reproducible study settings, not confirmed anonymous-farm hardware.

Each historical run is grouped under `outputs/full_experiments/gme_oracle/<run>/farm_<id>/<day>/` with `plots/`, `csv/`, `json/`, and optional `models/`. The effective configuration, input provenance and artifact manifest accompany auction plans, immutable ledger, actual dispatch, used PV vintages, settlement and objective comparisons. The main KPI distinctions are:

- `total_gain_eur`: actual feedback profit after available settlement;
- `perfect_information_objective_eur`: achieved full-information PV+BESS objective;
- `pv_only_perfect_information_objective_eur`: achieved optional PV-only objective;
- `battery_incremental_operating_value_eur`: the difference in (54);
- initial expectation, realized-minus-expected revenue/profit, and separate executed versus counterfactual degradation components.

Figures preserve interval widths and both available boundary points, including next-day local midnight. Held hourly/quarter-hour prices and powers do not create new samples; SOC uses the actual boundary states. Snapshot settlement diagnostics include their different delivery windows and common-horizon comparisons.

Solver settings use relative MIP gap 10^{-4} (0.01%) by default, with no absolute MIP gap configured. Objective comparisons use achieved `ObjVal`. Status, relative gap, model sizes and timing remain recorded. Before a schedule is used, independent validation recomputes physical balances, limits, native product consistency, information sharing, degradation and objective components. Execution separately checks actual SOC and power/POI protection. Software consistency does not resolve uncertain farm metadata, forecast bias or historical publication vintages.

13 Current limitations and extension boundaries

The current empirical workflow is a single-day perfect-price benchmark with one weather member, approximate geometry and an idealized interval-average controller. Measured AC is effective observed production, not an independently estimated unconstrained irradiance-to-generation potential; hidden plant curtailment/faults in source production are not identified automatically. Configured cleaning and fallback choices remain part of the experiment.

The model currently omits continuous trading, order acceptance, market impact, multiple-farm network coupling, ancillary services, intrainterval dynamics, ramping/startup constraints, dynamic capacity loss, calendar aging, thermal battery dynamics, fixed operating costs and investment cash-flow optimization. The generic `backtest` CLI remains synthetic-only; the separate historical `experiment-gme-oracle` workflow is the implemented manual-data replay. Future uncertain-price evaluations need their own causal inputs and comparison protocol rather than relabeling the current oracle results.

The original implementation-status report is retired. This source and its PDF are the maintained mathematical description; run manifests and saved solver results are the numerical evidence for particular configurations. Practical commands and configuration examples are in the root `README.md`.

References

[1] B. Xu, J. Zhao, T. Zheng, E. Litvinov and D. S. Kirschen, *Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*. Author preprint and published paper links retained by the repository: https://arxiv.org/abs/1707.04567 and https://doi.org/10.1109/TPWRS.2017.2733339.

[2] pvlib Python documentation. The installed library implements the solar-position, single-axis tracking, isotropic transposition, Faiman and PVWatts functions called by `forecasting/pv.py`. https://pvlib-python.readthedocs.io/.

[3] GME market timing references and period-specific study provenance are recorded in `markets.py` and the selected study's YAML. The January historical configuration explicitly labels assumed result-publication clocks.

[4] Repository source files listed in Section 1 and the effective configuration saved with each run are authoritative for implemented behavior. Conceptual references do not replace inspection of those inputs and policies.